

Goals for today

- **Admin items:**
 - Return to in-person modality starting Wednesday. **The new classroom is Gilmore 234.** I will still offer a synchronous Zoom option.
 - I'm reviewing project proposals and will provide feedback this week.
- Complete statics
- Introduce manipulability

Review of the Jacobian matrix

The i^{th} column of $J_s(\theta) \in \mathbb{R}^{6 \times n}$ is given by

$$J_{si}(\theta) = Ad_{e^{[S_1]\theta_1} \dots e^{[S_{i-1}]\theta_{i-1}}}(S_i) \text{ for } i = 2, \dots, n \text{ with the first column } J_{s1} = S_1$$

The i^{th} column of $J_b(\theta) \in \mathbb{R}^{6 \times n}$ is given by

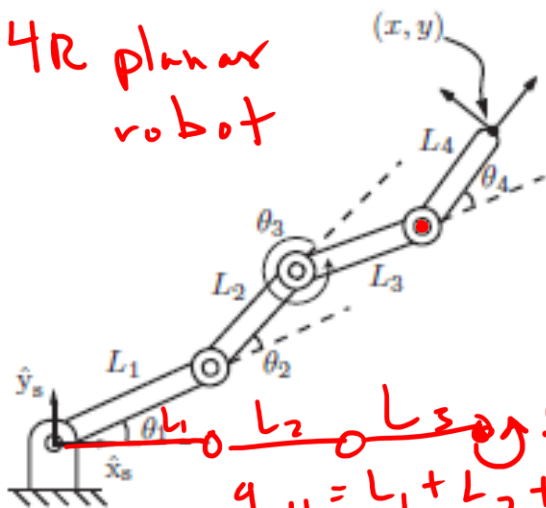
$$J_{bi}(\theta) = Ad_{e^{-[B_n]\theta_n} \dots e^{-[B_{i+1}]\theta_{i+1}}}(B_i) \text{ for } i = n - 1, \dots, 1 \text{ with } J_{bn} = B_n$$

The body and space Jacobians are related via

$$J_s(\theta) = [Ad_{T_{sb}}]J_b(\theta)$$

$$J_b(\theta) = [Ad_{T_{bs}}]J_s(\theta)$$

4R planar robot



For the forward kinematics, write down $M \in SE(2)$ and each $S_i = (\omega_{zi}, v_{xi}, v_{yi}) \in \mathbb{R}^3$

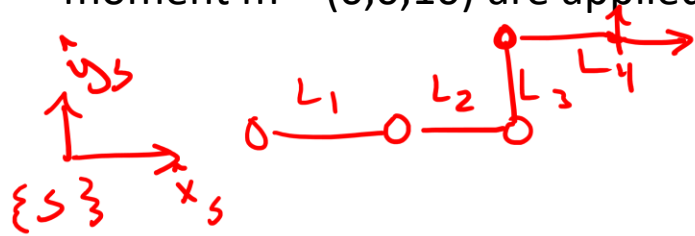
$$M = \begin{bmatrix} 1 & 0 & L_1 + L_2 + L_3 + L_4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{"zero" configuration}$$

$$\omega_{s_1} = \omega_{s_2} = \omega_{s_3} = \omega_{s_4} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow S_4 = \begin{bmatrix} \omega_{s_4} \\ v_{s_4} \end{bmatrix} = \begin{bmatrix} \omega_{s_4} \\ -\omega_{s_4} \times q_4 \end{bmatrix}$$

$$S_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -\omega_{s_4} \times q_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \times \begin{bmatrix} L_1 + L_2 + L_3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -(L_1 + L_2 + L_3) \\ 0 \end{bmatrix}$$

$$S = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -L_1 & -L_1 - L_2 & -L_1 - L_2 - L_3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Suppose that the chain is in static equilibrium at the configuration $[0, 0, \pi/2, -\pi/2]$ and that a force $f = (10, 10, 0)$ and a moment $m = (0, 0, 10)$ are applied to the tip (w.r.t the fixed frame). What are the torques experienced at each joint?



only s_4 changes

$$\omega_{s_4} = [0 \ 0 \ 1]^T$$

$$q_4 = [L_1 + L_2, L_3, 0]^T$$

$$v_{s_4} = -\omega_{s_4} \times q_4 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ L_1 + L_2 & L_3 & 0 \end{vmatrix} = \hat{i}(L_3) - \hat{j}(L_1 + L_2) + \hat{k}(0)$$

$$\text{Now, } J_s(\theta) = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & L_3 \\ 0 & 0 & 0 & L_3 \\ 0 & -L_1 & -L_1 - L_2 & -L_1 - L_2 \\ 0 & 0 & 0 & 0 \end{bmatrix}_{6 \times 4}$$

$$\tau_{4 \times 1} = J_s^T(\theta)_{4 \times 6} F_s_{6 \times 1}$$

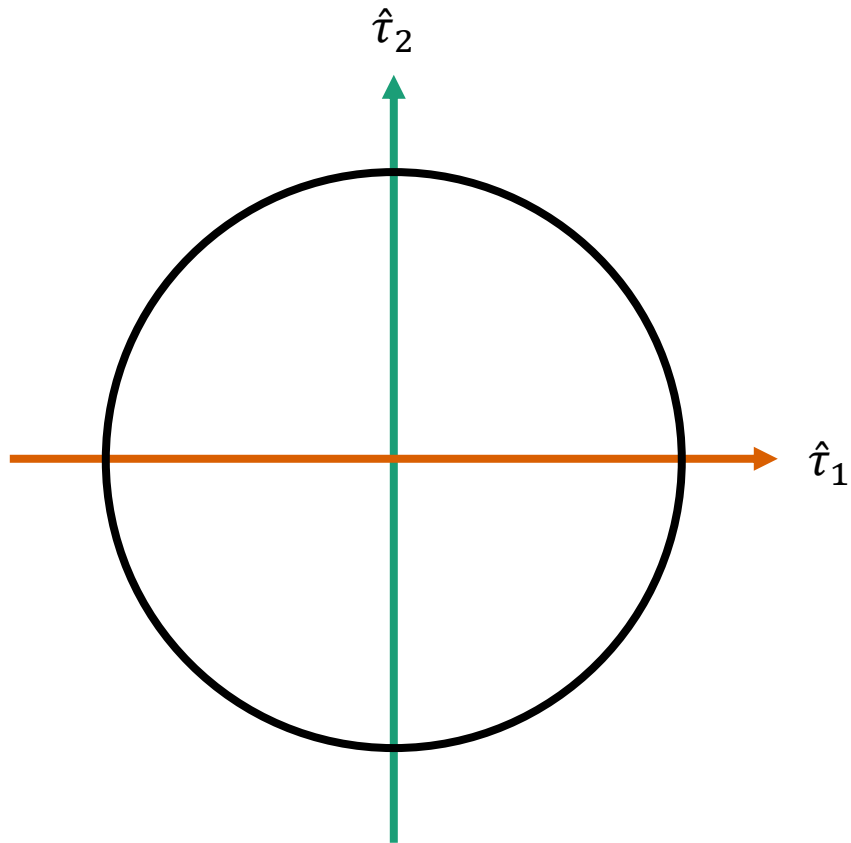
$$\text{and } \tau = J_s^T(\theta) F_s$$

$$\text{where } F_s = [0, 0, 10, 10, 10, 0]^T$$

$$\tau = 10 \begin{bmatrix} 2 \\ 1 - L_1 \\ 1 - L_1 - L_2 \\ 1 + L_3 - L_1 - L_2 \end{bmatrix} \text{ N.m}$$

Answer

Mapping joint torques to end-effector forces



Let the joint torques vary like so:

$$-1 \, N \cdot m \leq \tau_1 \leq 1 \, N \cdot m$$

$$-1 \, N \cdot m \leq \tau_2 \leq 1 \, N \cdot m$$

A unit circle, like that depicted on the left, represents an “iso-effort” force contour in the τ_1 - τ_2 plane.

Creating a force ellipsoid (1)

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -L_1 \sin(\theta_1) - L_2 \sin(\theta_1 + \theta_2) & -L_2 \sin(\theta_1 + \theta_2) \\ L_1 \cos(\theta_1) + L_2 \cos(\theta_1 + \theta_2) & +L_2 \cos(\theta_1 + \theta_2) \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

$\dot{x} =$

$J(\theta)$

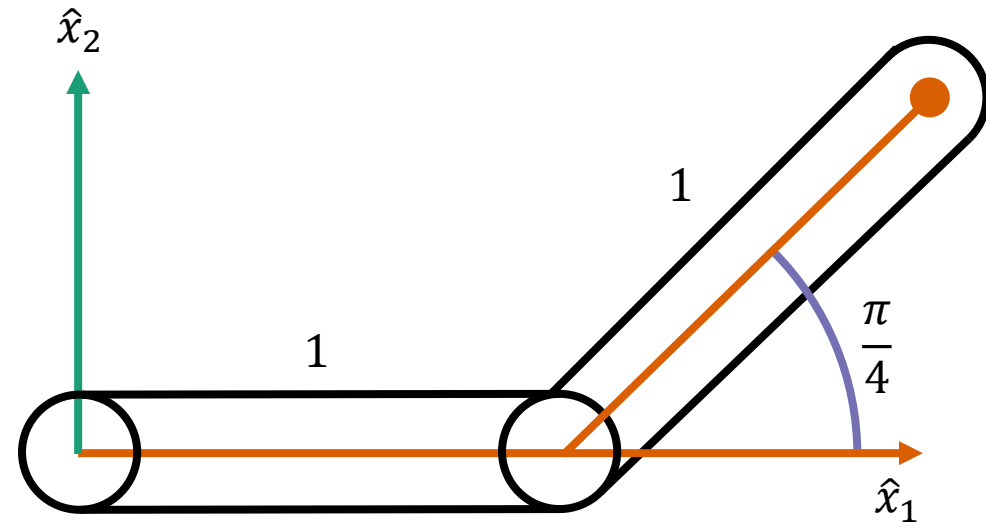
$\dot{\theta}$

Consider this planar, 2R arm. $\theta = [0, \pi/4]^T$

$$J\left(\begin{bmatrix} 0 \\ \pi/4 \end{bmatrix}\right) = \begin{bmatrix} -0.71 & -0.71 \\ 1.71 & 0.71 \end{bmatrix}$$

$$\tau = J^T(\theta)F \rightarrow F = J^{-T}(\theta)\tau$$

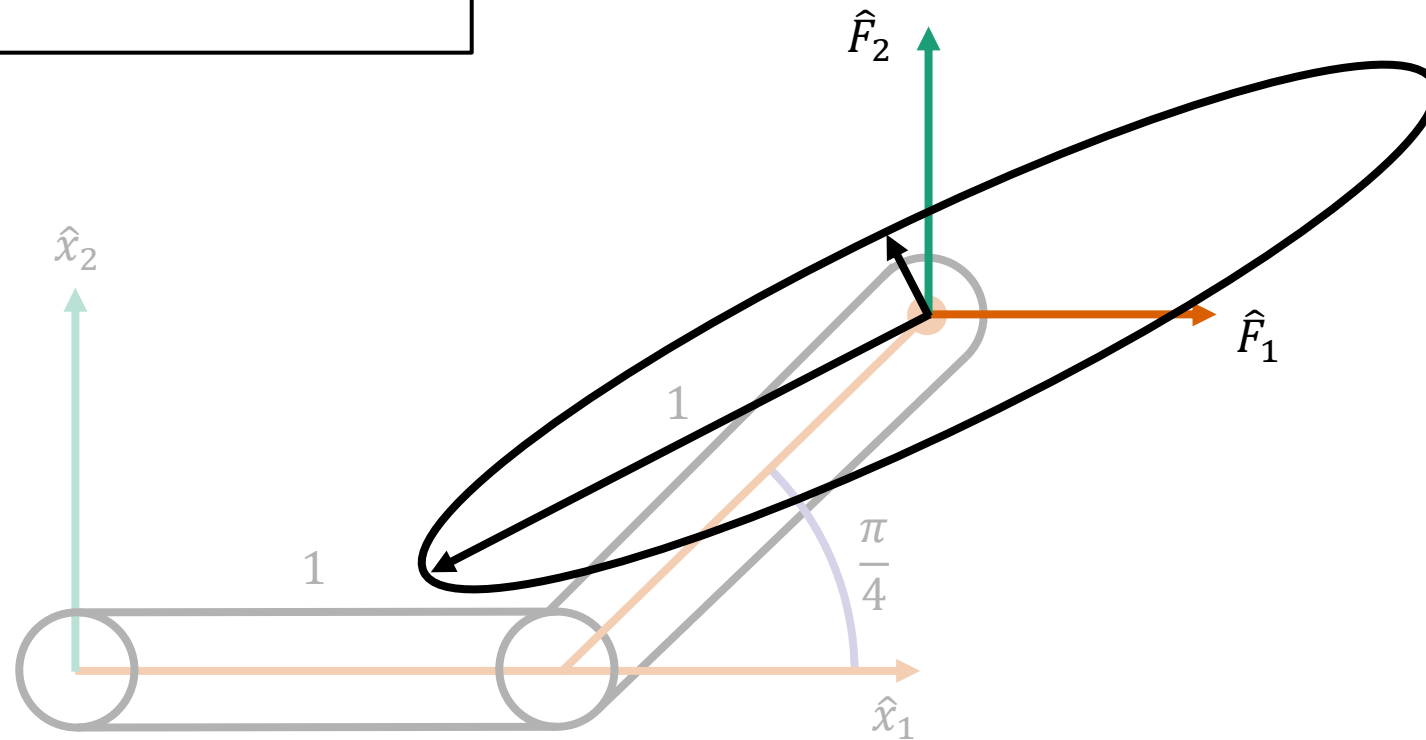
The **force ellipsoid** shows how easily the robot can generate forces in different directions at a particular posture.



Creating a force ellipsoid (2)

```
% Jacobian for a planar 2R arm (link lengths = 1)
J1 = [-0.71 -0.71; 1.71 0.71]; % theta = [0; pi/4]
% Force ellipsoid
angle = linspace(0,2*pi,100);
for i = 1:length(angle)
    tau(1,i) = 1*cos(angle(i));
    tau(2,i) = 1*sin(angle(i));
    F(:,i) = inv(J1')*tau;
end
plot(F(1,:),F(2,:))
```

*going around
the iso-force contour*



Review of linear algebra: Eigenvectors & eigenvalues

An **eigenvector** of a linear transformation is a nonzero vector that changes by a scalar factor when that linear transformation is applied to it. The corresponding **eigenvalue** is the factor by which the eigenvector is scaled.

Example:

$$A = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$$

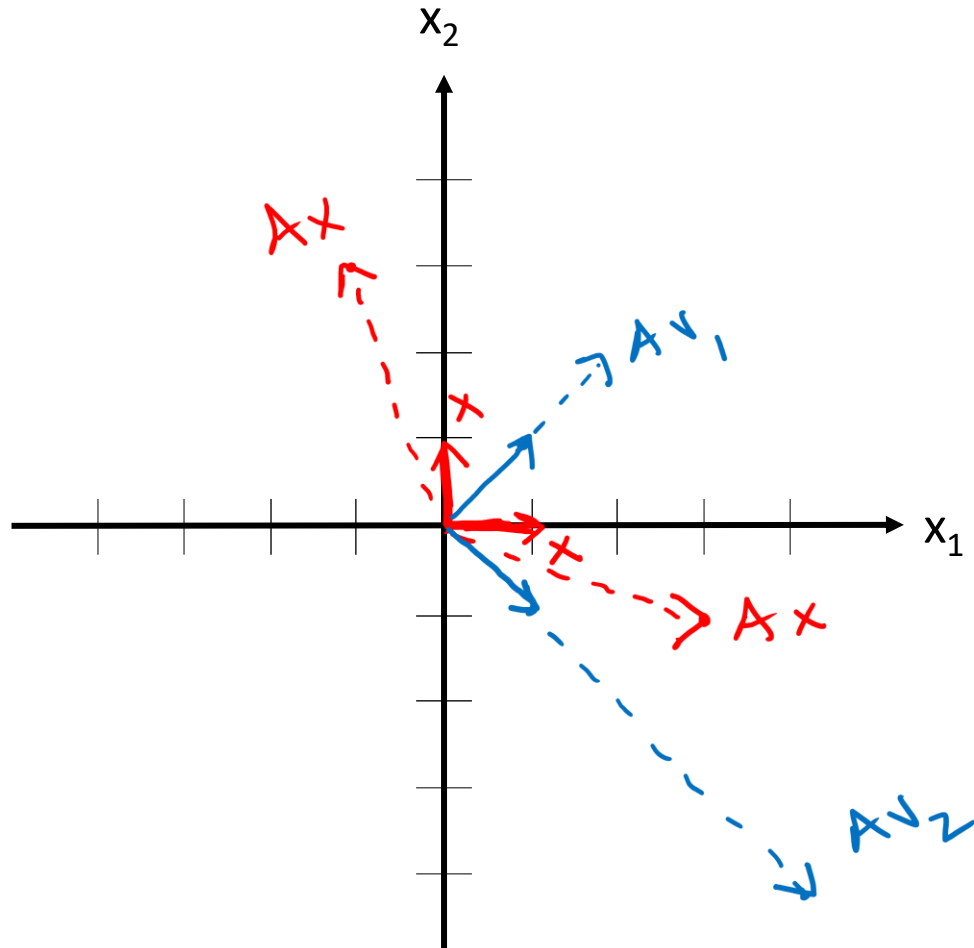
$$\text{try } x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow Ax = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \text{No! Not an eigenvector.}$$

$$\text{try } x = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow Ax = \begin{bmatrix} -1 \\ 3 \end{bmatrix} \quad \text{No!}$$

$$\text{try } v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow Av_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \quad \text{Yes!} \\ \lambda_1 = 2$$

$$\text{try } v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \rightarrow Av_2 = \begin{bmatrix} 4 \\ -4 \end{bmatrix} \quad \text{Yes!}$$

exactly 2 eigenvalues λ_1, λ_2
and 2 eigenvectors v_1, v_2
 $\lambda_2 = 4$



Eigenvalues & eigenvectors in general

$$Ax = \lambda x = \lambda Ix$$

↖ identity matrix

$$(A - \lambda I)x = 0$$

case 1 : $x = 0$ (the trivial solution ... boring!)

case 2 : $x \neq 0$ and $\det(A - \lambda I) = 0$

" $A - \lambda I$ " is singular, meaning that some vectors get mapped to zero

$$\boxed{\det(A - \lambda I) = 0}$$

characteristic
equ.

→ polynomial equation whose roots are the eigenvalues

Determinants

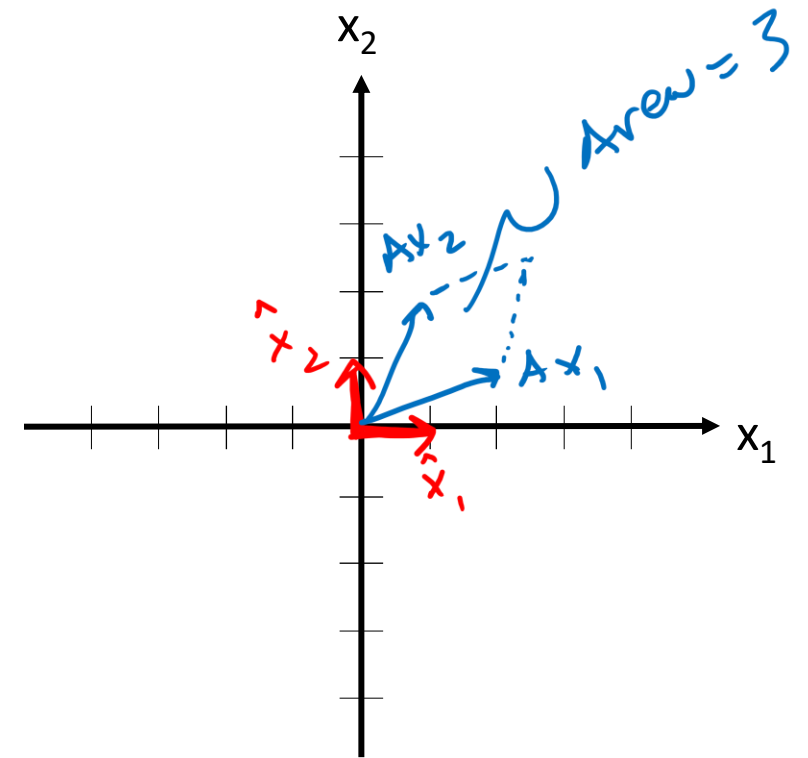
The determinant measures the volume of a unit cube after mapping through A

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\det(A) = 4 - 1 = 3 \checkmark$$

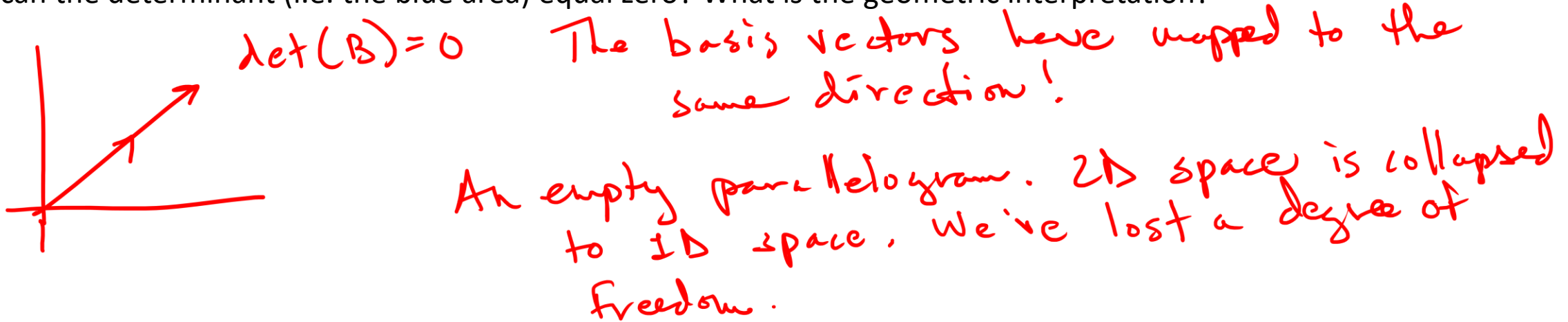
$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Determinants

How can the determinant (i.e. the blue area) equal zero? What is the geometric interpretation?



Higher dimensional space $\mathbb{R}^n$ goes to a lower dimensional space $\mathbb{R}^m$ if the determinant of the linear transformation is zero. Some vectors get mapped to zero.

$$[A - \lambda I]x = 0$$

the nonzero vectors that get mapped to zero if $[A - \lambda I]$ is singular

Singularities

We can also use the Jacobian to identify postures where the robot's end-effector loses the ability to move instantaneously in one or more directions. These postures are known as **singularities**.

At singularities, the Jacobian $J(\theta)$ is **less** than full rank. For a Jacobian $J(\theta) \in \mathbb{R}^{6 \times n}$, where n is the number of joints:

$$\text{rank}(J(\theta)) \leq \min(6, n)$$

- If $\text{rank}(J(\theta)) = \min(6, n)$, then $J(\theta)$ is full rank at that θ
- If $\text{rank}(J(\theta)) < \min(6, n)$, then $J(\theta)$ is **singular** at that θ

Some other definitions

For a Jacobian $J(\theta) \in \mathbb{R}^{6 \times n}$, where n is the number of joints:

- If $n < 6$: the system is **kinematically deficient** and $J(\theta)$ is “tall”
- If $n = 6$: $J(\theta)$ is “square”
- If $n > 6$: the system is **kinematically redundant** and $J(\theta)$ is “wide”

Singularities are independent of the choice of frame:

- $\text{rank}(J_s(\theta)) = \text{rank}(J_b(\theta))$

Singularities

Consider the body Jacobian for a 6R robot:

$$\mathcal{V}_b = [J_{b1}(\theta) J_{b2}(\theta) \cdots J_{b6}(\theta)] \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \vdots \\ \dot{\theta}_6 \end{bmatrix}$$

If, for example, columns 1 and 2 are linearly dependent:

$$\begin{aligned} \mathcal{V}_b &= J_{b1}(\theta)\dot{\theta}_1 + nJ_{b1}(\theta)\dot{\theta}_2 + \cdots + J_{b6}(\theta)\dot{\theta}_6 \\ &= J_{b1}(\theta)(\dot{\theta}_1 + n\dot{\theta}_2) + \cdots + J_{b6}(\theta)\dot{\theta}_6 \end{aligned}$$

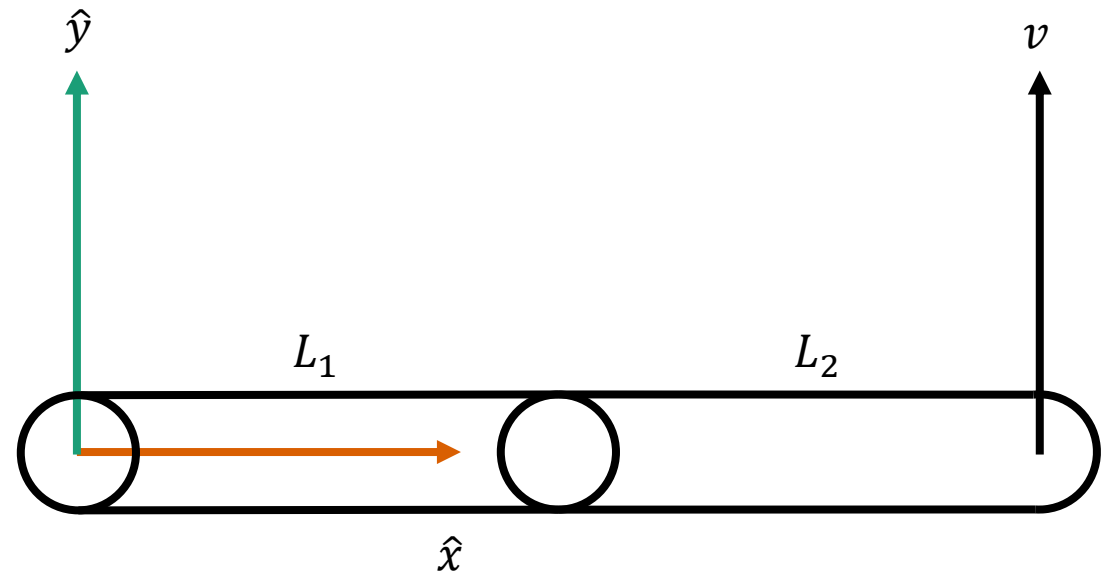
We lose the ability to generate instantaneous spatial velocity in one dimension

Planar 2R arm (1)

Consider the 2R arm where $\theta = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$J\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 0 & 0 \\ L_1 + L_2 & L_2 \end{bmatrix}$$

Since $\det(J(\theta)) = 0$, $J(\theta)$ is not full rank and is thus singular.



Planar 2R arm (2)

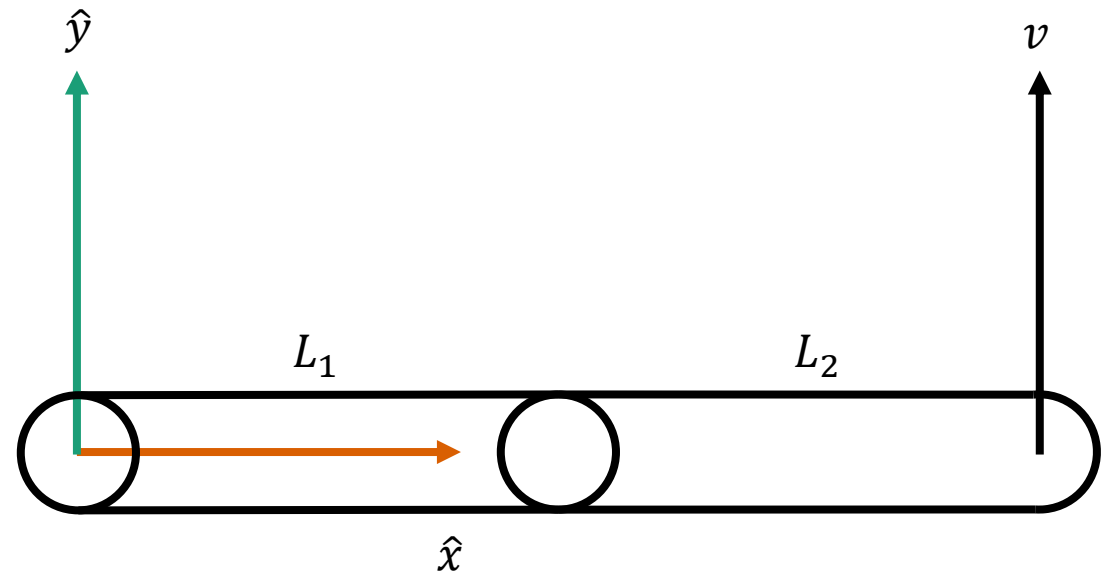
We can also see that any joint motion will cause end-effector motion in the $\hat{y}$ direction only.

To prove this:

$$\dot{x} = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ L_1 + L_2 & L_2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

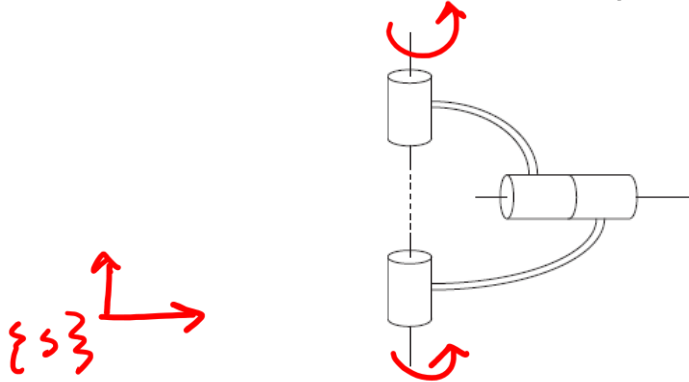
$$\dot{x} = 0; \dot{y} = (L_1 + L_2)\dot{\theta}_1 + L_2\dot{\theta}_2$$

Thus, even with 2 degrees of freedom, we can only instantaneously generate a tip velocity in the y-direction at this posture.



Some common singularities for 6R, serial chain robots

Two collinear revolute joint axes

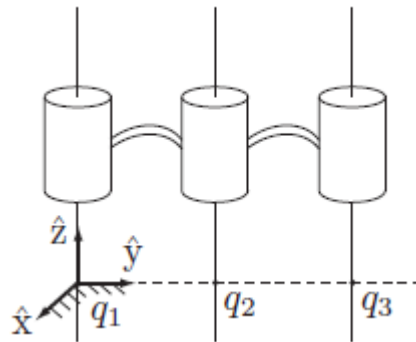


$$J_{s_1}(\theta) = \begin{bmatrix} \omega_{s_1} \\ -\omega_{s_1} \times q_1 \end{bmatrix} \quad \text{and} \quad J_{s_2}(\theta) = \begin{bmatrix} \omega_{s_2} \\ -\omega_{s_2} \times q_2 \end{bmatrix}$$

$$\omega_{s_1} = \omega_{s_2} \quad \text{and} \quad -\omega_{s_1} \times q_1 = -\omega_{s_2} \times q_2$$

$$J_s(\theta) = \begin{bmatrix} J_{s_1} & J_{s_1} & \cdots \end{bmatrix}$$

Three coplanar and parallel revolute joint axes



$$J_s(\theta) = \begin{bmatrix} \omega_{s_1} & \omega_{s_1} & \omega_{s_1} & \cdots \\ 0 & -\omega_{s_1} \times q_2 & -\omega_{s_1} \times q_3 & \cdots \end{bmatrix}$$

↑ we can write this column as a linear sum of the first 2 columns

For additional examples, see Sec. 5.3 of the textbook.

$$\begin{bmatrix} 5 & 5 & 5 & \cdots \\ 0 & 10 & 20 & \cdots \end{bmatrix} \quad \begin{matrix} A & B & C \end{matrix} \quad -A + 2B = \begin{bmatrix} 5 \\ 20 \end{bmatrix} = C$$

How close is the robot to a singularity?

Singularities are important to know because they limit the motion of our robot. However, a posture either is or is not a singularity. What about situations where a posture is very close to being a singularity?

We can use **manipulability ellipsoids** to see if a nonsingular configuration is close to being singular.

Manipulability ellipsoids (1)

For an n -joint open chain & task space with coordinates $q \in \mathbb{R}^m$ where $m \leq n$, let's consider the joint rates $\dot{\theta}$ where $\|\dot{\theta}\| = 1$ (a unit-sphere in n -dimensional joint-velocity space)

$$1 = \dot{\theta}^T \dot{\theta}$$

$$1 = (J^{-1} \dot{q})^T (J^{-1} \dot{q})$$

$$1 = \dot{q}^T J^{-T} J^{-1} \dot{q} = \dot{q}^T (JJ^T)^{-1} \dot{q}$$

$$1 = \dot{q}^T A^{-1} \dot{q} \text{ (a quadratic form)}$$

*This assumes J can be inverted

Manipulability ellipsoids (2)

If J is full rank, the matrix $A = JJ^T \in \mathbb{R}^{m \times m}$ is square, symmetric, and **positive definite**, as is A^{-1} .

Positive definite: a matrix A for which $x^T Ax > 0$ for all x . This also means that all eigenvalues λ of A are positive.

Positive semi-definite: a matrix A for which $x^T Ax \geq 0$ for all x . This also means that all eigenvalues λ of A are nonnegative.

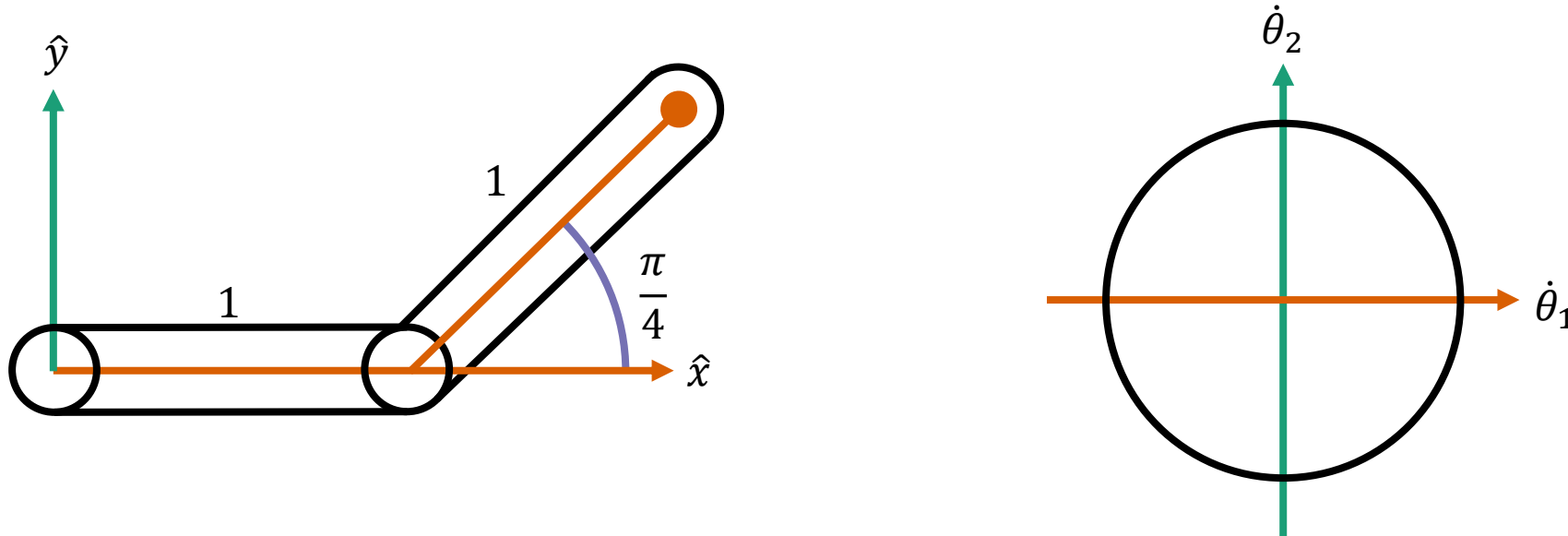
Manipulability ellipsoids (3)

Going back to $1 = \dot{q}^T A^{-1} \dot{q}$, the set of vectors $\dot{q}$ that satisfies this equation defines an ellipsoid in m -dimensional space. For this ellipsoid:

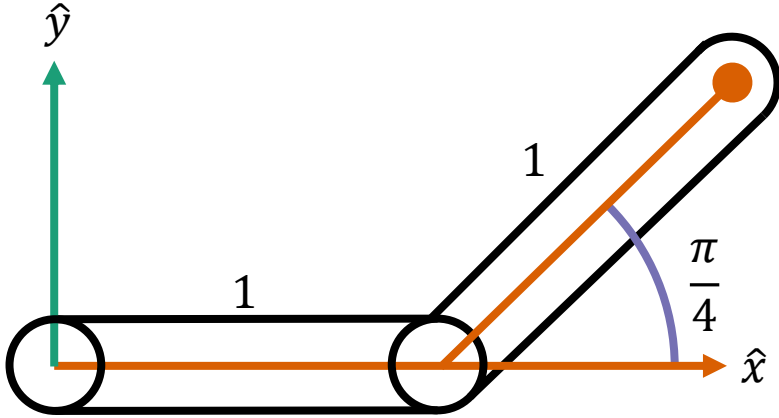
- $v_i \rightarrow$ eigenvectors of A , which represent the directions of the ellipsoid's principal semi-axes
- $\sqrt{\lambda_i} \rightarrow$ square roots of the eigenvalues of A , which represent the lengths of the ellipsoid's principal semi-axes

Manipulability ellipsoids: a simple example (1)

Let's consider the planar 2R arm again. The “iso-effort” contour in $\dot{\theta}_1$ - $\dot{\theta}_2$ space is shown below.



Manipulability ellipsoids: a simple example (2)



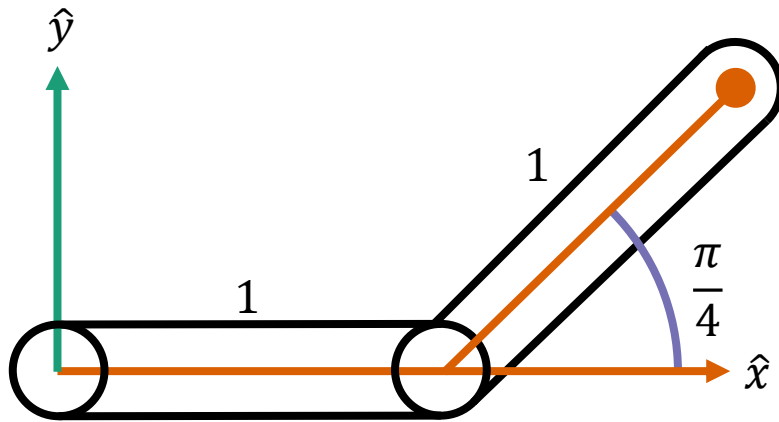
The Jacobian is:

$$J \left(\begin{bmatrix} 0 \\ \pi/4 \end{bmatrix} \right) = \begin{bmatrix} -0.71 & -0.71 \\ 1.71 & 0.71 \end{bmatrix}$$

We can therefore find:

$$A = JJ^T = \begin{bmatrix} 1.0082 & -1.7182 \\ -1.7182 & 3.4282 \end{bmatrix}$$

Manipulability ellipsoids: a simple example (3)

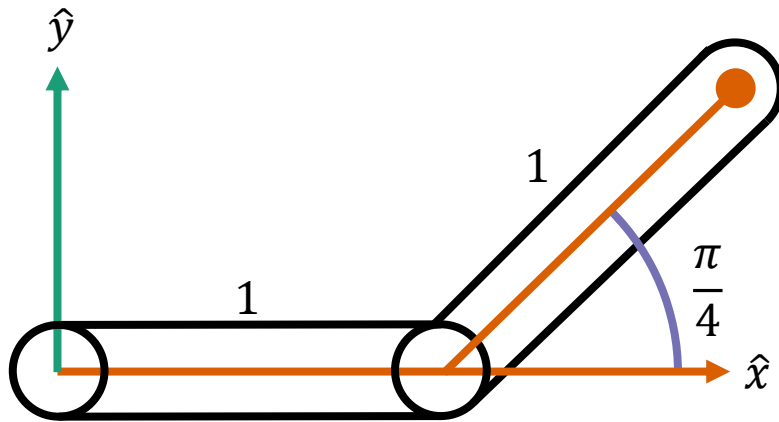


Let's find the eigenvalues. Starting from $Ax = \lambda x \rightarrow (A - \lambda I)x = 0$, we find the roots of the characteristic equation ($\det(A - \lambda I) = 0$) using Matlab or Python:

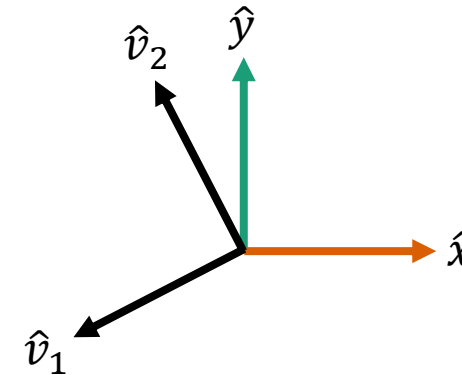
$$\lambda_1 = 0.1167, v_1 = \begin{bmatrix} -0.8876 \\ -0.4606 \end{bmatrix}$$

$$\lambda_2 = 4.3197, v_2 = \begin{bmatrix} -0.4606 \\ 0.8876 \end{bmatrix}$$

Manipulability ellipsoids: a simple example (4)



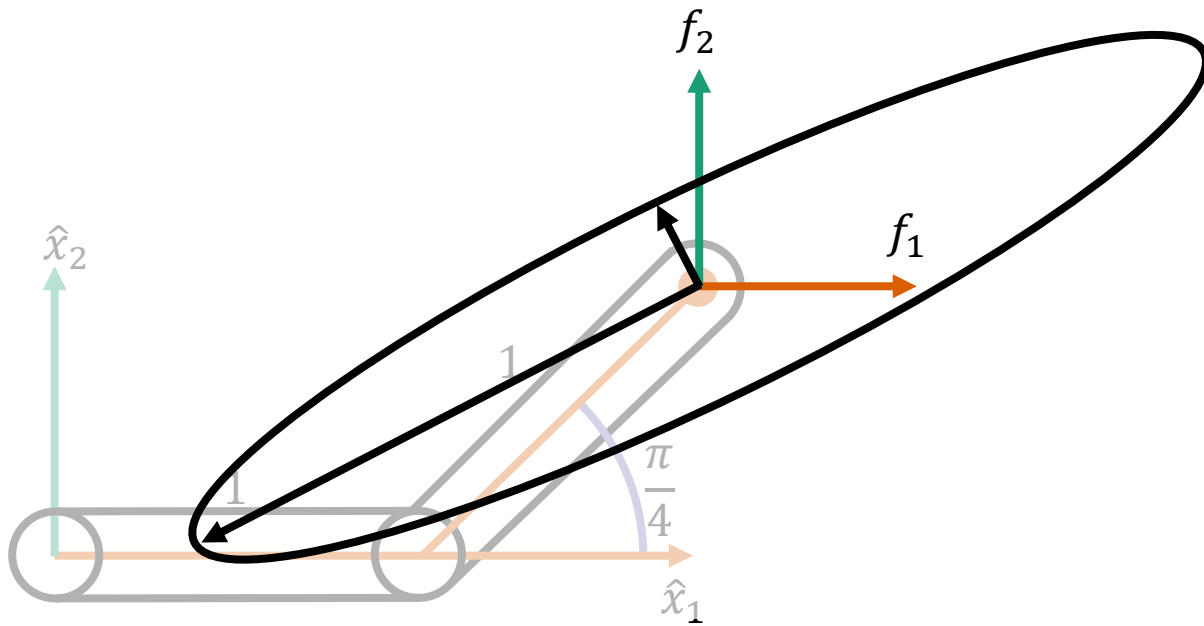
The manipulability ellipsoid's principal semi-axes thus look like:



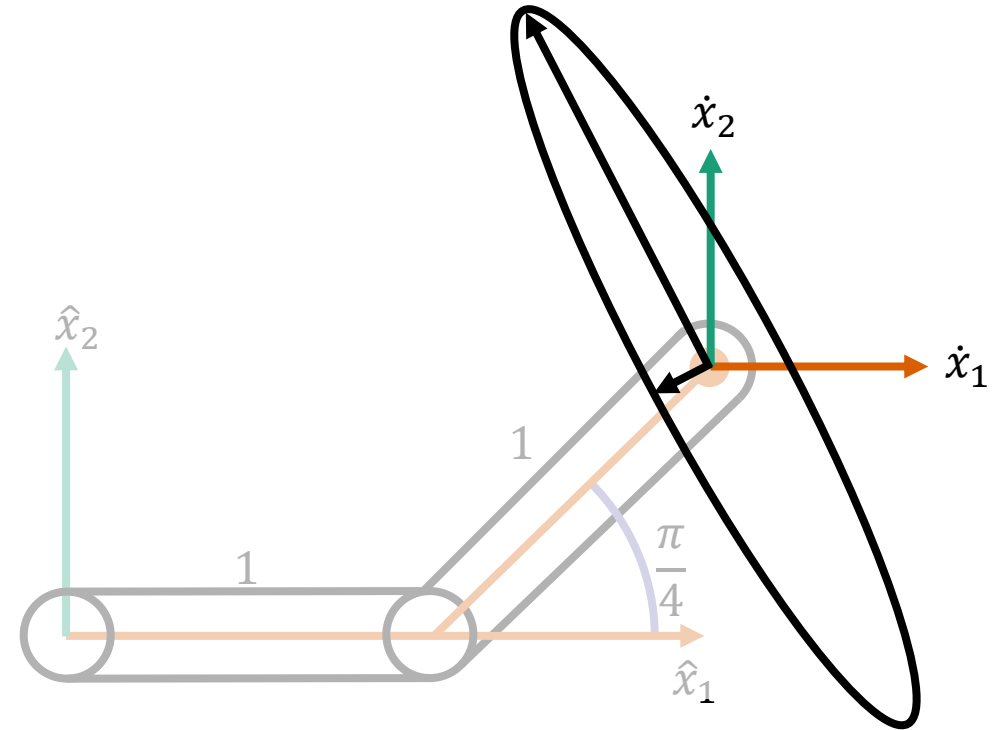
The unit axes $\hat{v}_1$ and $\hat{v}_2$ are then scaled by $\sqrt{0.1167}$ and $\sqrt{4.3197}$, respectively.

Manipulability ellipsoids: a simple example (5)

Force ellipsoid



Manipulability ellipsoid



Manipulability & force ellipsoids

The principal axes of manipulability ellipsoids and force ellipsoids are aligned (the eigenvectors of A are also the eigenvectors of $B = A^{-1}$).

The lengths of the principal semi-axes of the force ellipsoid are scaled by $1/\sqrt{\lambda_i}$ (rather than $\sqrt{\lambda_i}$ for the manipulability ellipsoid).

- If it's easy to generate a tip velocity in a certain direction, then it's difficult to generate a force in that direction (and vice versa)

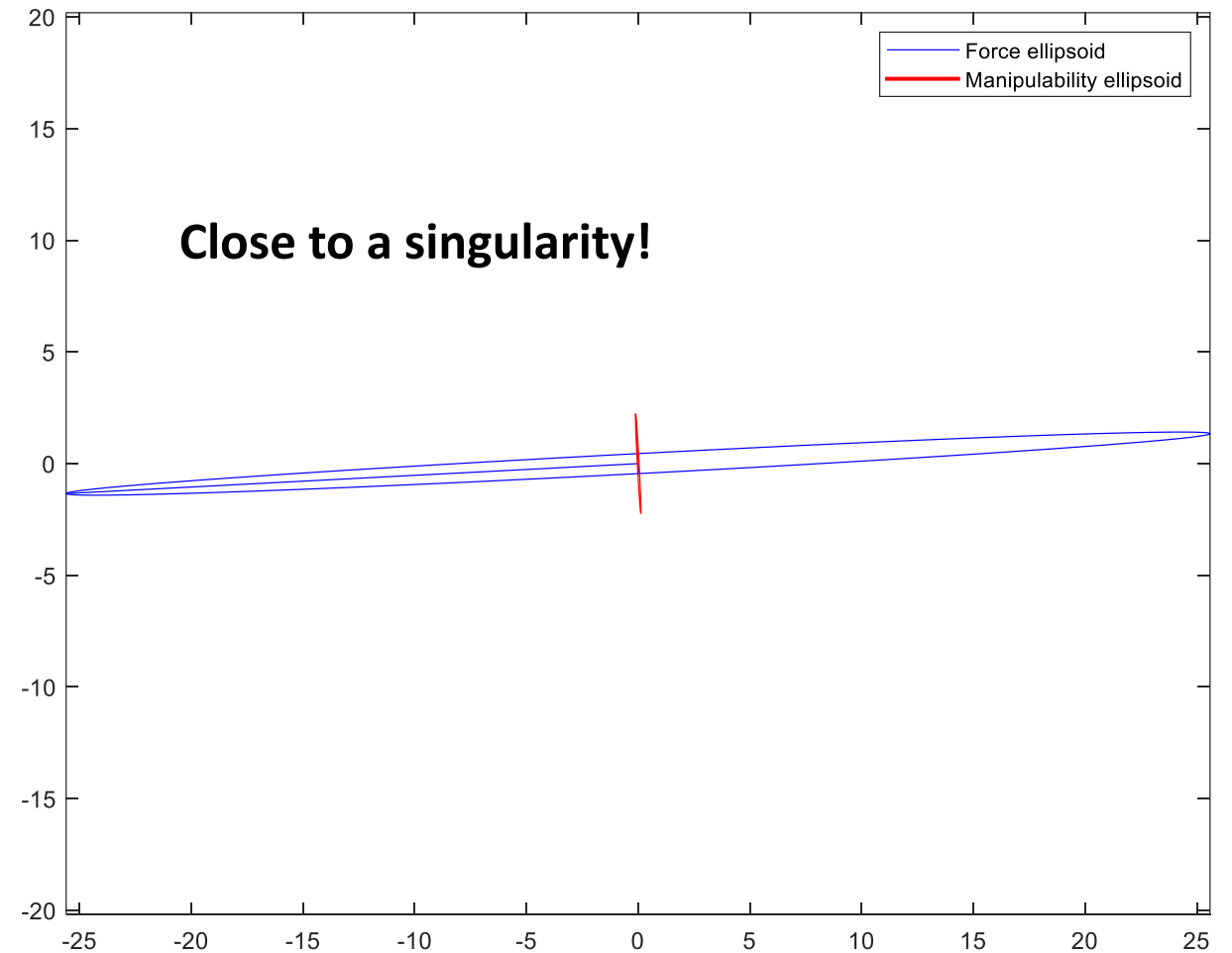
At a singularity:

- The manipulability ellipsoid collapses to a line segment
- The force ellipsoid becomes infinitely long orthogonal to the manipulability ellipsoid

```

% joint angles
theta = [0 5]; % degrees
% Jacobian for a planar 2R arm (link lengths = 1)
J1 = [-sind(theta(1))-sind(theta(1)+theta(2)), -sind(theta(1)+theta(2));
      cosd(theta(1))+cosd(theta(1)+theta(2)), cosd(theta(1)+theta(2))];
% Force ellipsoid
angle = linspace(0,2*pi,100);
for i = 1:length(angle)
    tau(1,1) = 1*cos(angle(i));
    tau(2,1) = 1*sin(angle(i));
    F(:,i) = inv(J1')*tau;
end
plot(F(1,:),F(2,:), 'b')
hold on
% Find principal semi-axes
[V,lamda] = eig((J1*J1')^-1);
plot([0,V(1,1)]*lamda(1,1)^0.5,[0,V(2,1)]*lamda(1,1)^0.5, 'b')
plot([0,V(1,2)]*lamda(2,2)^0.5,[0,V(2,2)]*lamda(2,2)^0.5, 'b')
% Manipulability ellipsoid
for i = 1:length(angle)
    tau(1,1) = 1*cos(angle(i));
    tau(2,1) = 1*sin(angle(i));
    Q(:,i) = J1*tau;
end
plot(Q(1,:),Q(2,:), 'r')
% Find principal semi-axes
[V,lamda] = eig(J1*J1');
plot([0,V(1,1)]*-lamda(1,1)^0.5,[0,V(2,1)]*-lamda(1,1)^0.5, 'r')
plot([0,V(1,2)]*lamda(2,2)^0.5,[0,V(2,2)]*lamda(2,2)^0.5, 'r')
axis equal
legend('Force ellipsoid', 'Manipulability ellipsoid')
hold off

```



0 ——— 0 ——— 1 — 5° = θ₂

Two different manipulability ellipsoids

We can separate the geometric Jacobian into two 3D manipulability ellipsoids, one for angular velocity and one for linear velocity:

$$J(\theta) = \begin{bmatrix} J_\omega(\theta) \\ J_v(\theta) \end{bmatrix}$$

Usually, it makes sense to use $J_b(\theta)$ for this.

Measures of manipulability ellipsoids

$$\mu_1(A) = \frac{\sqrt{\lambda_{\max}(A)}}{\sqrt{\lambda_{\min}(A)}}$$

When $\mu_1(A)$ is low, or close to 1, the ellipsoid is nearly spherical and the end-effector can move equally easily in any direction.

$$\mu_2(A) = \sqrt{\lambda_1 \lambda_2 \dots} = \sqrt{\det(A)}$$

The volume of the ellipsoid μ_2 is proportional to the product of the semi-axis lengths. A larger μ_2 indicates more overall manipulability, which is better!

The product of the volume of the manipulability ellipsoid $\mu_2(A)$ and the volume of the force ellipsoid $\mu_2(B)$ is constant and independent of θ ($V_A V_B = \text{constant}$).

The volume of one cannot increase without decreasing the volume of the other.

Looking ahead: Inverse velocity kinematics

$$\dot{x} = J(\theta)\dot{\theta} \quad \dot{\theta} = J^{-1}(\theta)\dot{x}$$

Say we want the tip of our 2R robot to move with a cartesian velocity $\dot{x} = [0.1 \ 0.1]^T \text{ m/sec}$

@ theta = [0, -45 degrees]

```
>> inv(J)*xd
```

```
ans =
```

```
0  
0.1414
```

@ theta = [0, -5 degrees]

```
ans =
```

```
-1.0430  
2.1904
```

@ theta = [0, -1 degrees]

```
ans =
```

```
-5.6290  
11.3589
```

@ theta = [0, -0.01 degrees]

```
ans =
```

```
1.0e+03 *  
  
-0.5729  
1.1458
```

